

Bayesian Inference

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Bayesian Stats

- Bayes' Theorem elements: Posterior, Likelihood, Prior (and Evidence).
- Review of the covid test problem
- From Prior probability to Prior PDF: Bernoulli-Beta
- Maximum A Posteriori Estimate and Credible Intervals
- Choosing Priors, Conjugacy
- Poisson-Gamma
- Predictive Posteriors

Bayes Theorem

Posterior



Likelihood



Prior



$$P(\theta|y)$$

=

$$\mathcal{L}(y|\theta) \pi(\theta)$$

$$\int \mathcal{L}(y|\theta) \pi(\theta) d\theta$$

Evidence



Workshop Example

Exercise 2.1: Bayes Theorem

You are given the following information:

1. In random testing, someone tests positive for a disease.
2. In 5% of cases where the subject does not have the disease, this test shows positive anyway: i.e. there is a 5% false positive rate
3. There are no false negatives.
4. In the population at large, one person in a thousand has the disease.

What is the probability that the person tested actually has the disease?

Data Labels and Outcome Values

Hypothesis	Negative	Positive
Sick	<i>False Negative</i> $p = 0$	<i>True Positive</i> $p = 1$
Healthy	<i>True Negative</i> $p = 0.95$	<i>False Positive</i> $p = 0.05$

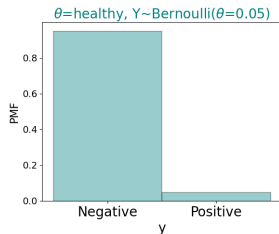
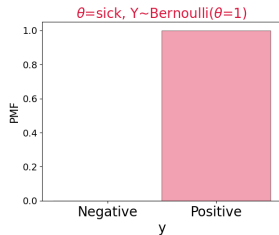
Probabilities

Data are Bernoulli - binary outcome of single test.

Hypothesis	Negative	Positive
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Sick	$p = 0$	$p = 1$
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Healthy	$p = 0.95$	$p = 0.05$
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Recap: $Y \sim \text{Bernoulli}(\theta)$

$$p_Y(y|\theta) = \theta^y (1 - \theta)^{1-y}$$

$$\mathcal{L}(y|\theta) = p_Y(y|\theta) \text{ for } N=1$$

PMF for sick people has $\theta_+ = 1$:

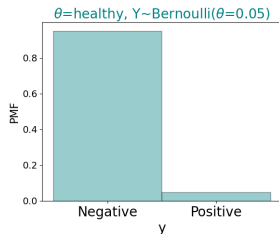
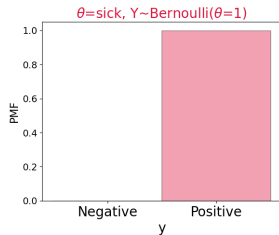
$$\mathcal{L}(y_+|\theta_+) = 1^1(0)^0 = 1$$

$$\mathcal{L}(y_-|\theta_+) = 1 - \mathcal{L}(y_+|\theta_+) = 0$$

PMF for healthy people has $\theta_- = 0.05$:

$$\mathcal{L}(y_+|\theta_-) = 0.05^1(0.95)^0 = 0.05$$

$$\mathcal{L}(y_-|\theta_-) = 1 - \mathcal{L}(y_+|\theta_-) = 0.95$$

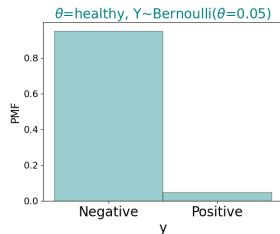
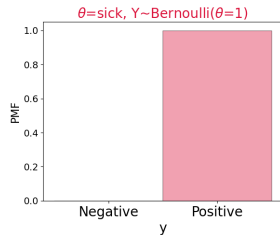


Likelihood Table for positive tests

Hypothesis **Likelihood of y_+**

$$\theta_+ \quad \mathcal{L}(y_+|\theta_+) = 1$$

$$\theta_- \quad \mathcal{L}(y_+|\theta_-) = 0.05$$



Likelihood Table for positive tests

Hypothesis	Likelihood	Prior	Bayes Numerator	Evidence	Posterior
θ	$\mathcal{L}(y_+ \theta)$	$\pi(\theta)$	$\mathcal{L}(y_+ \theta)\pi(\theta)$	$\int \mathcal{L}\pi d\theta$	$P(\theta y_+)$
θ_+	1	1/1000	0.001	0.05095	0.0196
θ_-	0.05	999/1000	0.04995	0.05095	0.9804

We calculated the Posteriors using Bayes' Theorem:

$$P(\theta_+|y_+) = \frac{\mathcal{L}(y_+|\theta_+) \pi(\theta_+)}{\int \mathcal{L}(y_+|\theta) \pi(\theta) d\theta} = \frac{\mathcal{L}(y_+|\theta_+) \pi(\theta_+)}{\mathcal{L}(y_+|\theta_+) \pi(\theta_+) + \mathcal{L}(y_+|\theta_-) \pi(\theta_-)}$$

Bayesian Updating

What we did in the workshop is called Bayesian Updating.

We updated our estimate of the probability from the *Prior*, $\pi(\theta)$, to the *Posterior*, $P(\theta|y)$.

We did this using the data y , and the Likelihoods $\mathcal{L}(y|\theta)$ provided to us by the test manufacturer for the two possible hypotheses θ .

- Before looking at the data, we had a Prior $\pi(\theta_+) = 0.001$.
- After the test we calculated a Posterior $P(\theta_+|y_+) = 0.0196 \approx 0.02$.

Cowboy alert! Results are meaningless without uncertainties...

Evidence

Recall that we need the Evidence (Bayes denominator) to ensure the Posterior is a valid PDF: $\int P(\theta|y) d\theta = 1$.

$$P(\theta_+|y_+) = \frac{\mathcal{L}(y_+|\theta_+) \pi(\theta_+)}{\int \mathcal{L}(y_+|\theta) \pi(\theta) d\theta} = \mathcal{L}(y_+|\theta_+) \pi(\theta_+) \times \mathbf{c}$$

However, it is just a **constant factor**, identical for every hypothesis, so we could **make life easier for ourselves**:

- $P(\theta_+|y_+) = \mathcal{L}(y_+|\theta_+) \pi(\theta_+) \mathbf{c} = 1 \times 0.001 \mathbf{c} = 0.001 \mathbf{c}$
- $P(\theta_-|y_+) = \mathcal{L}(y_-|\theta_+) \pi(\theta_+) \mathbf{c} = 0.05 \times 0.999 \mathbf{c} = 0.04995 \mathbf{c}$
- $\int P(\theta|y_+) d\theta = 0.05095 \mathbf{c} = 1 \quad \therefore \mathbf{c} \approx 19.627$

This shortcut gives us **Posteriors** of (0.0196, 0.9804) as before.

This theory



$$P(\theta|y)$$

=

This theory



$$\mathcal{L}(y|\theta)$$

$$\pi(\theta)$$

← This theory

$$\int \mathcal{L}(y|\theta) \pi(\theta) d\theta$$



All possible theories

This theory



$$P(\theta|y)$$

This theory



$$\mathcal{L}(y|\theta) \pi(\theta)$$

← This theory

$$\int \mathcal{L}(y|\theta) \pi(\theta) d\theta$$

Normalisation
constant



$$c = \int P(\theta|y) d\theta$$

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

$$P(\theta|y) \propto \mathcal{L}(y|\theta) \pi(\theta)$$

Form of Prior

For our Priors, we assigned $\pi(\theta_-) = 999/1000$ and $\pi(\theta_+) = 1/1000$.

This was a very specific choice, made as though we could know with absolute certainty the rate of cases at that given time, which is of course impossible.

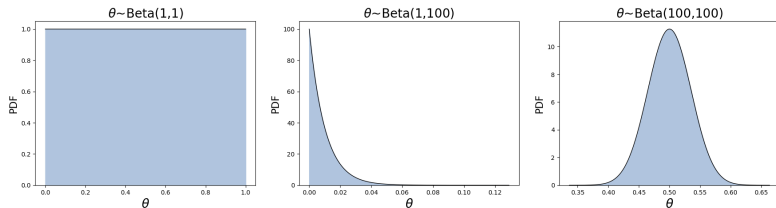
The Prior probability should be a PDF* with some specific distribution in the range $[0,1]$. This is what the **Beta(α, β)** distribution is made for.

* If the prior is not a valid PDF, it is an 'Improper Prior' - these can still be used.

Recap: $Y \sim \text{Beta}(\alpha, \beta)$

$$f_{\theta}(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Recall the $\text{Beta}(\alpha, \beta)$ distribution is a chameleon, being able to take on a huge range of shapes depending on its two shape hyper-parameters α, β .



The $B(\alpha, \beta)$ in the PDF ensures normalisation.

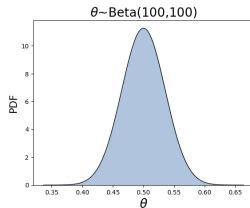
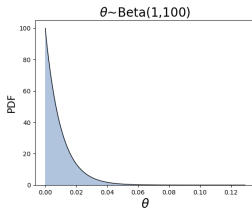
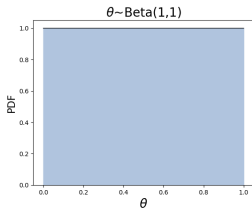
`from scipy.special import beta as betafunc` (same name as the `scipy.stats` beta distribution - careful!):

`betafunc(1,1)=1`; `betafunc(0.1, 99.9)=6.006` etcetera.

Recap: $Y \sim \text{Beta}(\alpha, \beta)$

$$f_{\theta}(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Recall the $\text{Beta}(\alpha, \beta)$ distribution is a chameleon, being able to take on a huge range of shapes depending on its two shape hyper-parameters α, β .



PDF: $f_{\theta}(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$ with $0 \leq \theta \leq 1$, $\alpha > 0$, $\beta > 0$

Expected Value:

$$E[\theta] = \frac{\alpha}{\alpha + \beta}$$

and Variance:

$$V[\theta] = \frac{E[\theta](1 - E[\theta])}{(\alpha + \beta + 1)}$$

The Posterior $P(\theta|y) \propto \mathcal{L}(y|\theta) \pi(\theta)$

Beta Priors: $\pi(\theta) \propto \theta^{\alpha-1} (1-\theta)^{\beta-1}$

Bernoulli Likelihoods: $\mathcal{L}(y|\theta) = \theta^y (1-\theta)^{1-y}$ (for N=1 data point)

The Posterior : $P(\theta|y) \propto \mathcal{L}(y|\theta) \pi(\theta)$

$$\propto \theta^y (1-\theta)^{1-y} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$$\propto \theta^y \theta^{\alpha-1} (1-\theta)^{1-y} (1-\theta)^{\beta-1}$$

$$\propto \theta^{y+\alpha-1} (1-\theta)^{1-y+\beta-1}$$

$$\propto \theta^{(\alpha+[y])-1} (1-\theta)^{(\beta+[1-y])-1}$$

$$\propto \theta^{\alpha'-1} (1-\theta)^{\beta'-1}$$

The Posterior $P(\theta|y) \propto \text{Beta}(\alpha', \beta')$ where $\alpha' = \alpha + [y]$ and $\beta' = \beta + [1-y]$.

Bernoulli and Beta are Conjugate: **Beta = Bernoulli $\times$ Beta**.

The Posterior

Priors: $\pi(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$

Posteriors: $P(\theta|y) \propto \theta^{(\alpha+[y])-1} (1-\theta)^{(\beta+[1-y])-1}$

If $y = 0$: $P(\theta|y=0) \propto \theta^{\alpha-1} (1-\theta)^{[\beta+1]-1}$

⇒ A negative result increments $\beta \rightarrow [\beta + 1]$ wrt prior.

If $y = 1$: $P(\theta|y=1) \propto \theta^{[\alpha+1]-1} (1-\theta)^{\beta-1}$

⇒ A positive result increments $\alpha \rightarrow [\alpha + 1]$ wrt prior.

The Posterior Expectation for the Covid Test

We cannot use the formula for the Posterior directly with $\theta = 1$ and $y = 1$, as it would give us a zero posterior for any value of $\beta \neq 1$.

$$P(\theta|y) \propto \theta^{(\alpha+y)-1} (1-\theta)^{(\beta-y+1)-1}$$

$$P(\theta = 1|y = 1) \propto 1^\alpha (0)^{\beta-1} \quad \Rightarrow \text{only non-zero for } \beta = 1.$$

Could we use a *Beta*($\alpha, 1$)?

$$\text{We want } E[\theta] = \frac{\alpha}{\alpha + \beta} = \frac{\alpha}{\alpha + 1} = 0.001$$

A value $\alpha = 0.001$ satisfies this. Let's have a look at the distribution.

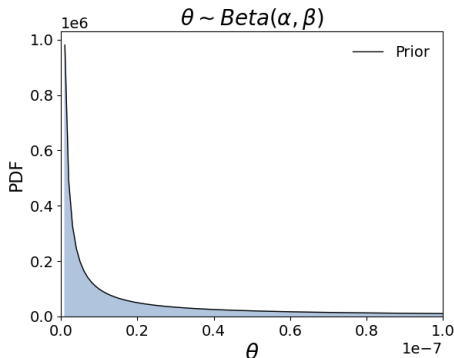
The Posterior Expectation for the Covid Test

The $Beta(0.001, 1)$ is almost entirely concentrated at values below 10^{-7} .

The tail extends to 1, enabling the expectation $E[\theta] = 10^{-3}$.

The Posterior expectation is

$$E[P] = \frac{\alpha + 1}{\alpha + 1 + \beta} = \frac{1.001}{2.001} \approx 0.5$$



This is not a distribution I would choose to model prevalence of a disease!

All Models Are Wrong (not just these. All.)

The **Frequentist** model is that the posterior is the likelihood.

$$P(\theta_+|y_+) = \mathcal{L}(y_+|\theta_+) = 100\%.$$

With a **Single-valued Prior** of 1/1000, the posterior is

$$P(\theta_+|y_+) = 0.0196 \approx 2\%.$$

This prior model suggests infinite precision and zero uncertainty, so is worrisome.

With a (lazy, crazy) **Beta Prior** the posterior is $P(\theta_+|y_+) \approx 50\%$.

This prior model gave us $E[\theta_+] = 0.001$ and a nonzero uncertainty, but clearly wasn't "the one".

Which model is the 'wrongest'?

Other choices we could have made

Insisting on an uncertainty means we must have a distribution, not a number, for our prior. Where did we get the 1/1000 prior? What are our sources of uncertainty?

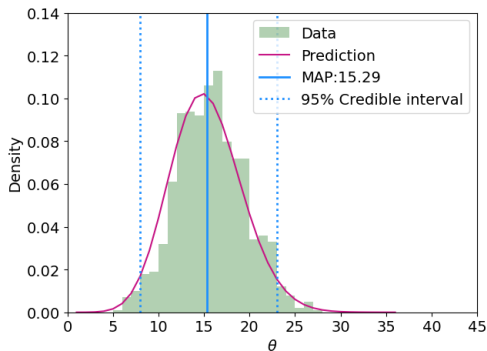
The ONS have a very good description of their [methodology](#) and a [github repo](#) with all the data and the analysis code. This is absolutely what everyone must always do if they want their analysis to be taken seriously.

It seems they used a Bayesian Posterior, and the prevalences quoted to the public were **Maximum A Posteriori** (MAP) estimates. I [downloaded the data](#) and saw that around April 2021, the MAP prevalence in England was 0.1% with a 95% **Credible Interval** of (0.08, 0.12)%

Based on this ten mins of research, I have adjusted my guess for the Prior: I would try a Normal (non-conjugate) Prior with mean 0.1% (as before) and sigma 0.02%. A non-conjugate prior means we would have to solve numerically.

Maximum A Posteriori Estimation

The value of θ that maximises the Posterior distribution (the mode of the posterior) is the MAP estimate.



$$\text{Gamma mode} = \frac{\alpha' - 1}{\beta'}$$

$$\text{Beta mode} = \frac{\alpha' - 1}{\alpha' + \beta' + 2}$$

α', β' are hyper-parameters of the Posterior distribution.

Maximum A Posteriori Estimation

$$P(\theta|y) \propto \mathcal{L}(y|\theta) \pi(\theta)$$

$$\arg \max_{\theta} P(\theta|y) = \arg \max_{\theta} \mathcal{L}(y|\theta) \pi(\theta)$$

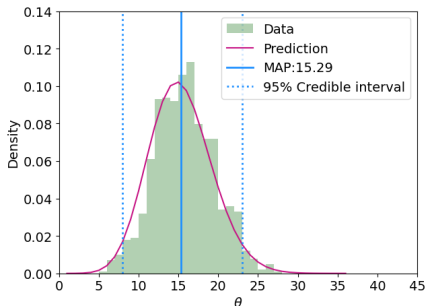
If the Prior $\pi(\theta)$ is uniform, the $\hat{\theta}_{MAP}$ is equivalent to the $\hat{\theta}_{MLE}$.

The difference between $\hat{\theta}_{MAP}$ and $\hat{\theta}_{MLE}$ is that the former uses not just data, but also our prior beliefs on the parameter's value.

In the limit of large datasets, $\hat{\theta}_{MAP} \rightarrow \hat{\theta}_{MLE}$. The main feature (until we look at MCMC magic) of Bayesian Posterior updating is its formidable performance with tiny amounts of data.

Credible Intervals

A **Credible Interval** is a range of values of the posterior that contain the true parameter with a certain probability, for example 95%. This intuitive definition is what many people incorrectly think the (unintuitive) Confidence Interval* is.



*A **Confidence Interval** is a range of values that contain the true parameter with a certain frequency. If you repeat the experiment 100 times you expect to find the value in the confidence interval 95 times.

Notes on Priors

Why do we need a Prior?

A Frequentist would say we don't, and may object to the use of a prior on the basis that it is a guess and is subjective. They would say $P(\theta|y) = \mathcal{L}(y|\theta)$.

A Bayesian might respond by pointing out that sampling distributions are also subjective guesswork.

Why did we choose a Beta Prior?

I briefly mentioned that [Beta and Bernoulli are Conjugate](#). This means that if I use a Beta Prior with a Bernoulli Likelihood, I get a Beta Posterior. Having a conjugate prior makes it possible to express the Posterior in closed form, and make it easier to see how the data updates the parameters.

Notes on Priors

If my Prior belief is just a number, can I use that?

I can't think of a situation where you would have a prior with no uncertainty (infinite precision). Better to use a distribution with a central value equal to your number, and variance equal to the square of your uncertainty. And check the distribution doesn't look insane.

Why if I don't have any Prior belief?

If you have no insight into the prior, you can use an [Uninformative Prior](#), or a very weakly informative prior. Usually this is a Uniform distribution on the real number line (∞ integral: an Improper Prior). As long as the Posterior is proper, this is okay. See [\[Jeffrey's Prior\]](#). Take care - a Prior that is Uniform in X is not Uniform in a transform of X .

Notes on Priors

What if my prior is a constraint?

If you have a constraint, for example if your parameter must physically be positive, but can theoretically have values $(-\infty, \infty)$, it may be tempting to use a uniform prior with a hard boundary at zero. This should be avoided, because strange things happen near hard boundaries. It is better to go with a soft turn on rather than a hard constraint.

Okay. Should I always use a Beta Prior, then?

Beta is a good choice if your parameter is a probability $(0,1)$.

If your parameter is a rate or count $(0,\infty)$, then the Gamma is the Conjugate Prior. Sometimes a Normal Prior is most appropriate. See [\[wikipedia\]](#) for a table of conjugate priors.

Conjugate Priors: Beta

```
scipy.stats.beta(a=alpha, b=beta)
```

Beta is conjugate for probability outcomes ($0 \leq \theta \leq 1$) such as Bernoulli, Binomial, Geometric

Beta priors: $\pi(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1} \Leftrightarrow \text{Beta}(\alpha, \beta)$

Bernoulli: $p_Y(y|\theta) = \theta^y(1-\theta)^{1-y}$

Likelihood: $\mathcal{L}(y|\theta) = \theta^y(1-\theta)^{1-y}$

Posterior: $P(\theta|y) \propto \theta^{(\alpha+y)-1}(1-\theta)^{(\beta-y)-1} \Leftrightarrow \text{Beta}(\alpha+y, \beta-y)$

Binomial: $p_Y(y|\theta) = C_y^n \theta^y(1-\theta)^{n-y}$

Likelihood: $\mathcal{L}(y|\theta) \propto \theta^y(1-\theta)^{n-y}$

Posterior: $P(\theta|y) \propto \theta^{(\alpha+y)-1}(1-\theta)^{(\beta+n-y)-1} \Leftrightarrow \text{Beta}(\alpha+y, \beta+n-y)$

Conjugate Priors: Gamma

```
scipy.stats.gamma(a=alpha, scale=1/beta)
```

Gamma is conjugate is for counting data ($0 \leq \theta \leq \infty$): Poisson, Exponential.

Gamma priors: $\pi(\theta) \propto \theta^{\alpha-1} e^{-\beta\theta} \Leftrightarrow \text{Gamma}(\alpha, \beta)$

Poisson: $p_Y(y|\theta) = e^{-\theta} \frac{\theta^y}{y!}$

Likelihood: $\mathcal{L}(y|\theta) \propto \theta^{n\bar{y}} e^{-n\theta}$

Posterior: $P(\theta|y) \propto \theta^{(\alpha+n\bar{y})-1} e^{-(\beta+n)\theta} \Leftrightarrow \text{Gamma}(\alpha + n\bar{y}, \beta + n)$

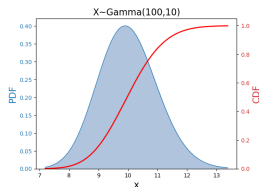
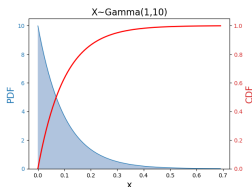
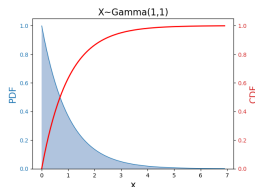
Exponential: $f_Y(y|\theta) = \theta e^{-\theta y}$

Likelihood: $\mathcal{L}(y|\theta) = \theta^n e^{-n\theta\bar{y}}$

Posterior: $P(\theta|y) \propto \theta^\alpha e^{-(\beta+n\bar{y})\theta} \Leftrightarrow \text{Gamma}(\alpha + 1, \beta + n\bar{y})$

The Gamma PDF

$$f(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}$$



The α is a shape hyper-parameter, and the β is a scale.

The **Expected Value** is $E[\theta] = \alpha\beta$ and the **Variance** is $V[\theta] = \alpha\beta^2$

Poisson-Gamma

Gamma priors: $\pi(\theta) \propto \theta^{\alpha-1} e^{-\beta\theta}$

Poisson likelihoods: $\mathcal{L}(y|\theta) \propto \theta^{n\bar{y}} e^{-n\theta}$

Gamma Posterior: $\text{Gamma}(\alpha + n\bar{y}, \beta + n)$

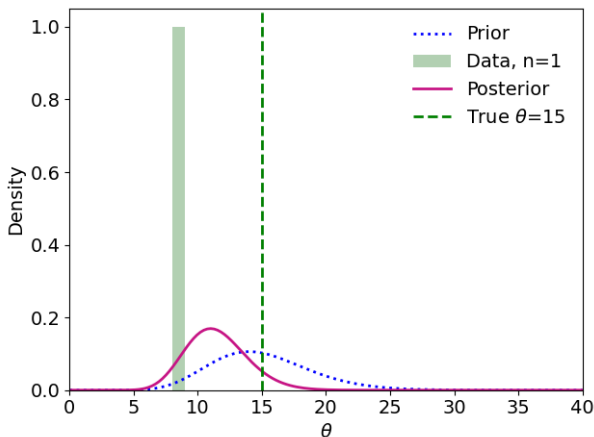
Note that for Gamma, the Expected Value is $E[\theta] = \alpha\beta$ and the Variance is $V[\theta] = \alpha\beta^2$

Compare with Poisson $E[y] = \theta$ and $V[y] = \theta$

For example, if we have Poisson data and guess at the rate parameter $\theta = 15$, we want $\alpha\beta = 15$ and $\alpha\beta^2 = 15$ ie $\alpha = 15, \beta = 1$.

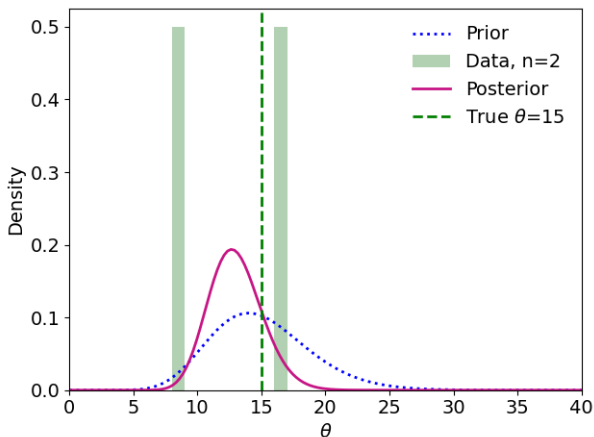
Poisson-Gamma: $\text{Gamma}(\alpha + n\bar{y}, \beta + n) \propto \mathcal{L}(\text{Data} | \theta = 15) \times \text{Gamma}(\alpha, \beta)$

The static prior is $\text{Gamma}(15, 1)$. True mean is $\theta = 15$.



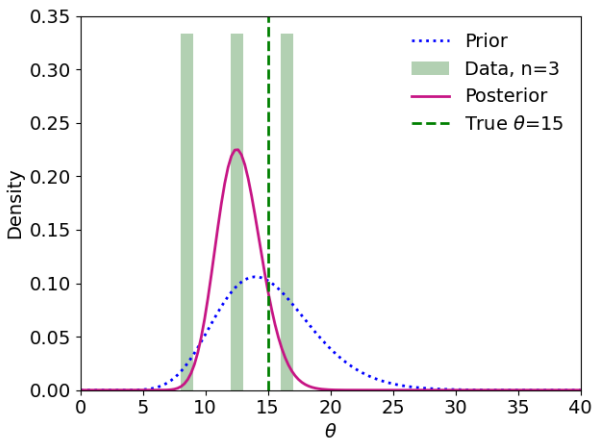
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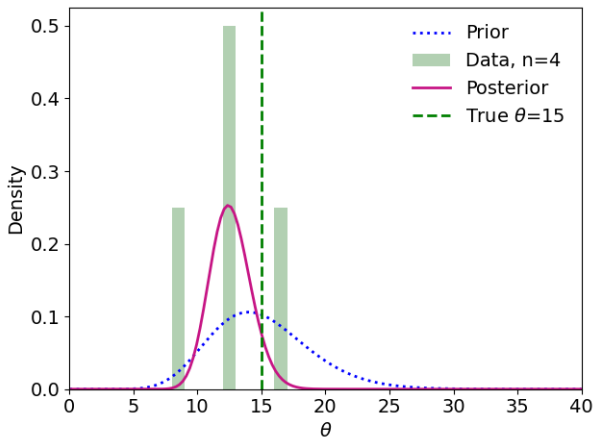
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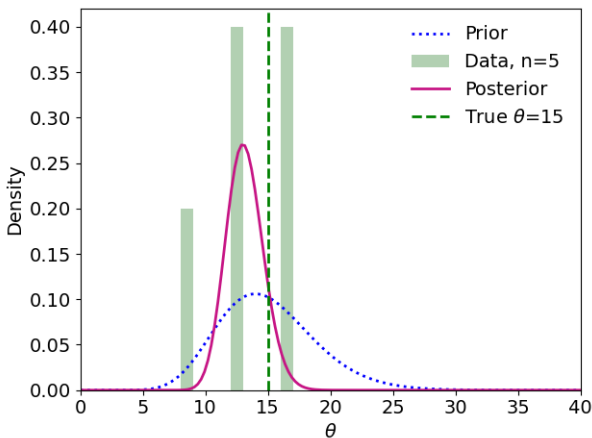
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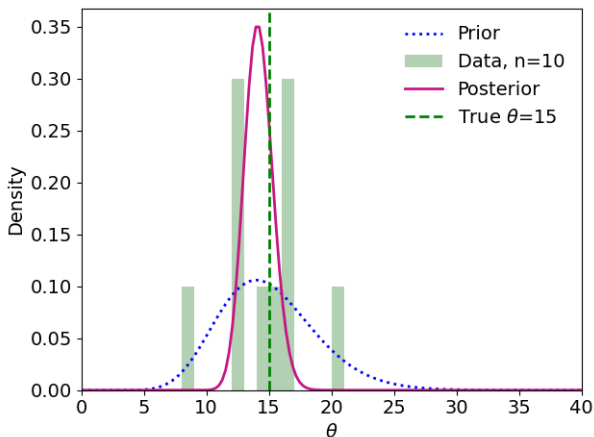
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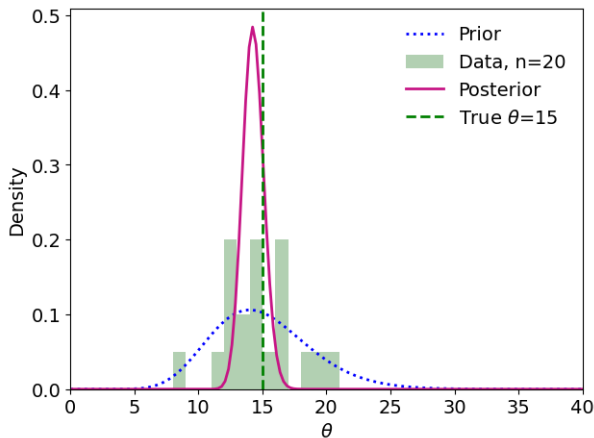
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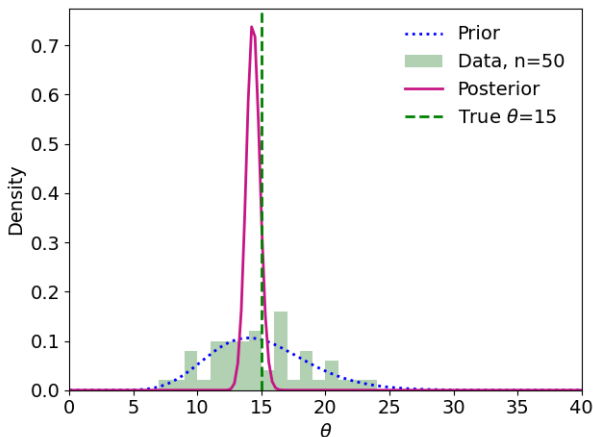
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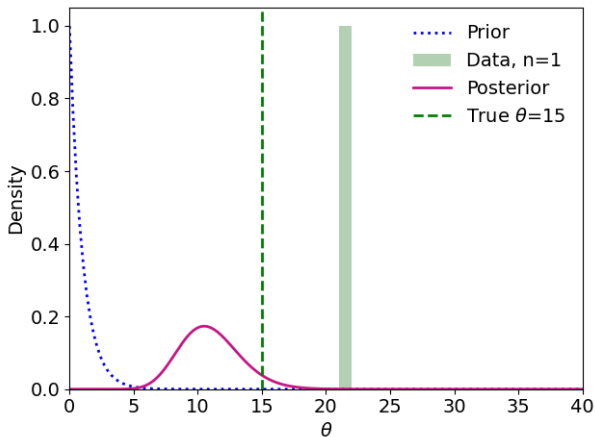
Poisson-Gamma: $\text{Gamma}(\alpha + n\bar{y}, \beta + n) \propto \mathcal{L}(\text{Data} | \theta = 15) \times \text{Gamma}(\alpha, \beta)$

The static prior is $\text{Gamma}(15, 1)$. True mean is $\theta = 15$.



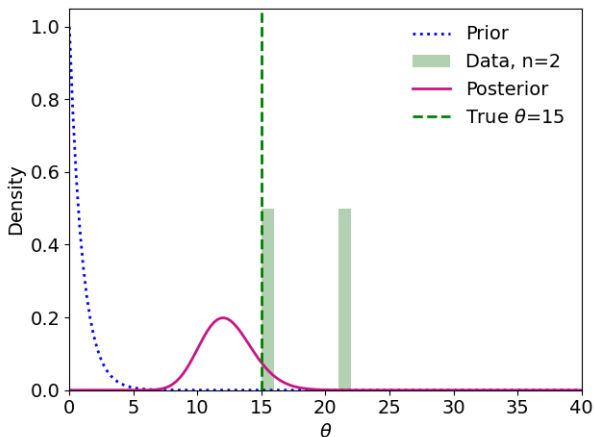
Poisson-Gamma

What if we start with a really bad guess: $\text{Gamma}(1,1)$?:



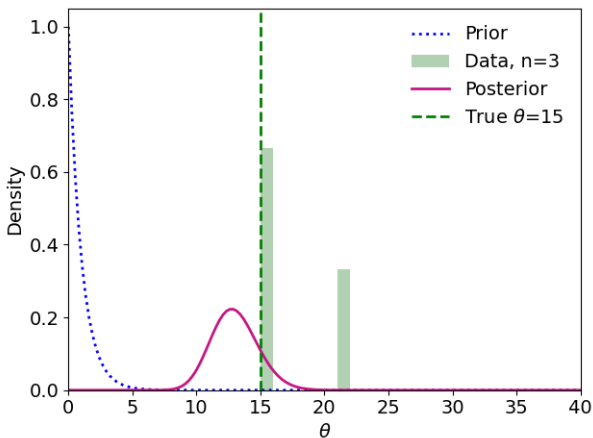
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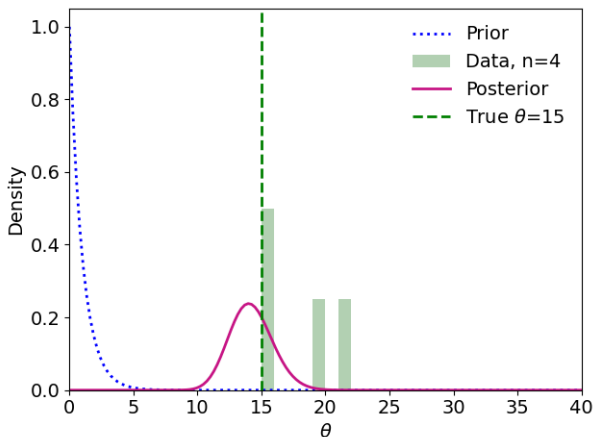
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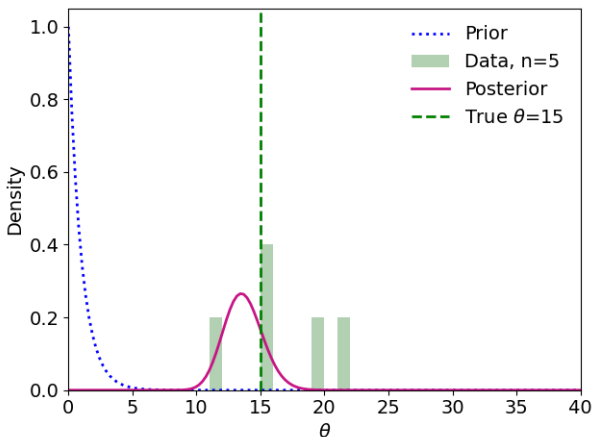
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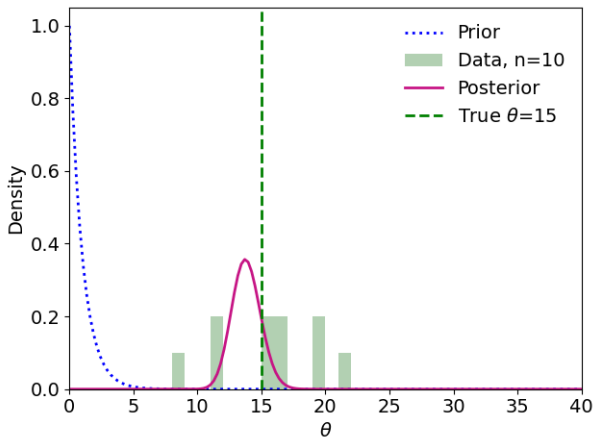
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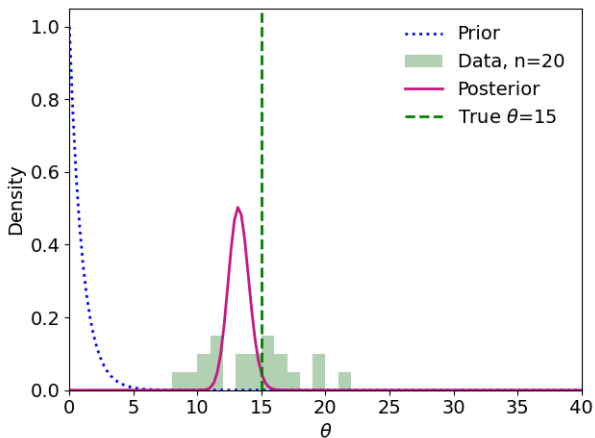
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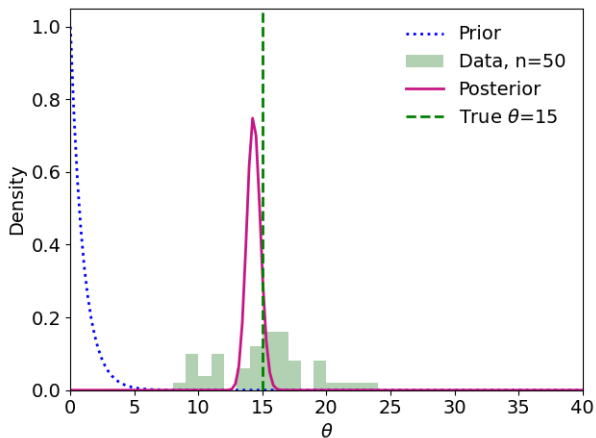
Poisson-Gamma

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Poisson-Gamma

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Predictions

We have seen that we can use Bayes Updating to find a Posterior Probability that will give us a pretty good estimate of the mean of a distribution.

The ingredients of the posterior are the Prior: what sort of distribution shape do we expect? and the Data itself.

Helping the Data along with a Prior allows the Posterior to work well with even just a few events.

If our prior is a bad guess, the posterior will take a little longer to converge, but quite quickly the information in the data overrides any mistakes in the prior.

It would be so cool if we could predict the whole distribution, rather than just the mean. Which we can!

Predictions

- We have seen that we can use Bayes Updating to find a Posterior Probability that will give us a pretty good estimate of the mean of a distribution.
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- Helping the Data along with a Prior allows the Posterior to work well with even just a few events.
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- It would be so cool if we could predict the whole distribution, rather than just the mean. And we can!

Predictive Posterior

↓

$$P(\tilde{y}|y) = \int_{\Theta} f(\tilde{y}|\theta) P(\theta|y) d\theta$$

Integrate out θ

PDF or PMF

Posterior

$$P(\theta|y) \propto \mathcal{L}(y|\theta) \pi(\theta)$$

Predictive Posterior

The (Marginal) **Predictive Posterior** describes the probability of a future measurement $\tilde{y}$, given our existing **pdf** and **posterior**, is:

$$P(\tilde{y}|y) = \int_{\Theta} f(\tilde{y}|\theta) P(\theta|y) d\theta$$

For Poisson-Gamma:
$$P(\tilde{y}|y) = \int_0^{\infty} e^{-\theta} \frac{\theta^{\tilde{y}}}{\tilde{y}!} \frac{\beta'^{\alpha'}}{\Gamma(\alpha')} \theta^{\alpha'-1} e^{-\beta'\theta} d\theta,$$

where the hyper-parameters of the **Posterior Gamma** are $\alpha' = \alpha + n\bar{y}$ and $\beta' = \beta + n$

The resulting Predictive $P(\tilde{y}|y)$ is a Negative Binomial:

$$NBinom\left(\alpha + n\bar{y}, \frac{\beta + n}{(\beta + n + 1)}\right)$$

Predictive Posterior

```
prediction = nbinom.pmf(xvals, n=n, p=p )
```

The Negative Binomial is just the Binomial, but for measuring failures rather than successes.

- Neg Binomial: $p_Y(y) \propto \theta^n (1 - \theta)^y$ models the number of failures y . n is the number of successes, θ is the probability of one success.
- Binomial: $p_Y(y) \propto \theta^y (1 - \theta)^{n-y}$ where y is the number of successes and n is the total.

As a predictive prior for Poisson-Gamma, the parameters are:

$$n = \alpha + n\bar{y} \text{ and } p = \frac{\beta + n}{(\beta + n + 1)}$$

Poisson-Gamma

Gamma priors: $\pi(\lambda) \propto \theta^{\alpha-1} e^{-\beta\theta}$

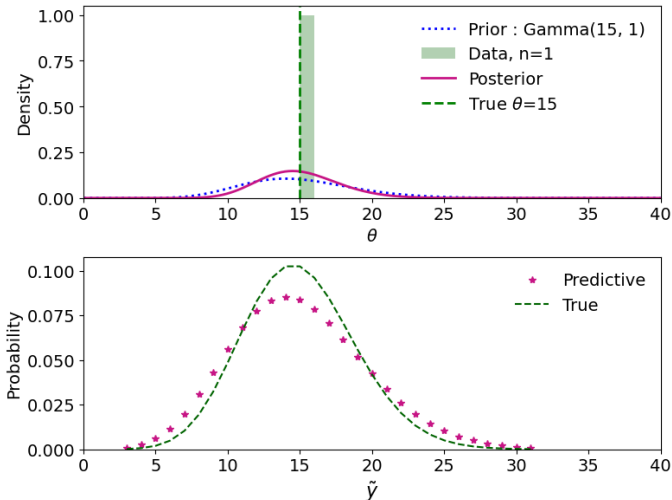
Poisson likelihoods: $\mathcal{L}(y|\theta) \propto \theta^{n\bar{y}} e^{-n\theta}$

Gamma Posterior: $\text{Gamma}(\alpha + n\bar{y}, \beta + n)$

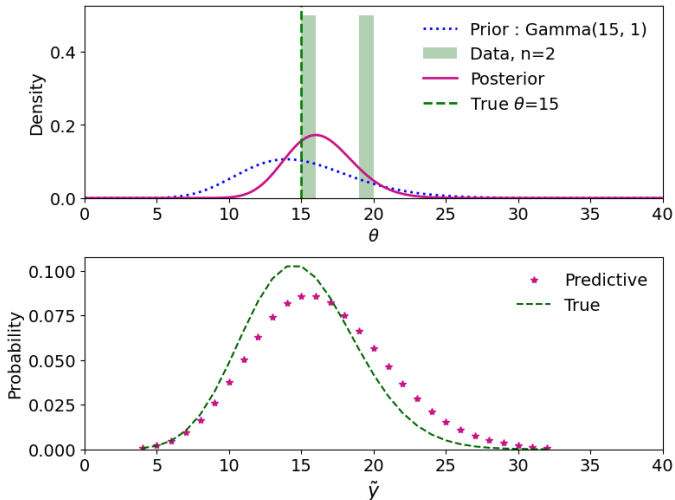
Neg Binom Predictive Posterior: $\text{NBinom}(\alpha + n\bar{y}, (\beta + n)/(\beta + n + 1))$

The Predictive Posterior is pretty neat, because unlike the Posterior which shrinks in variance around its mean value as the data increases, the predictive posterior remains the same shape as the data.

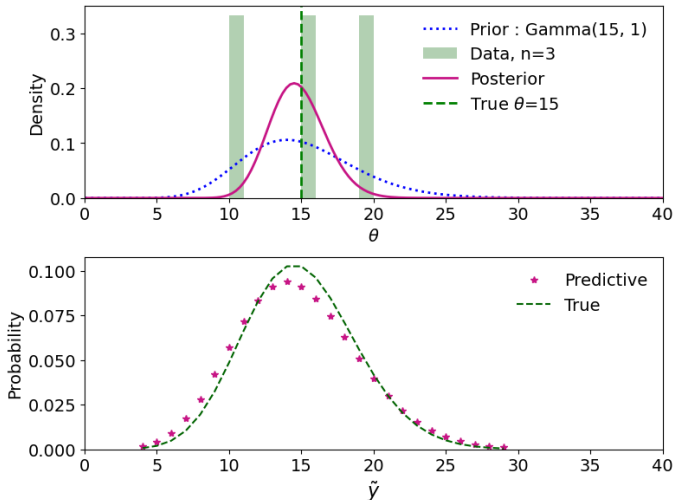
Predictive Posterior



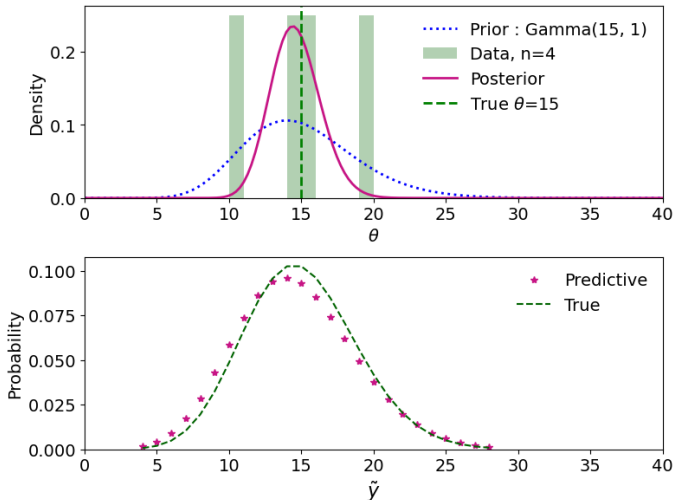
Predictive Posterior



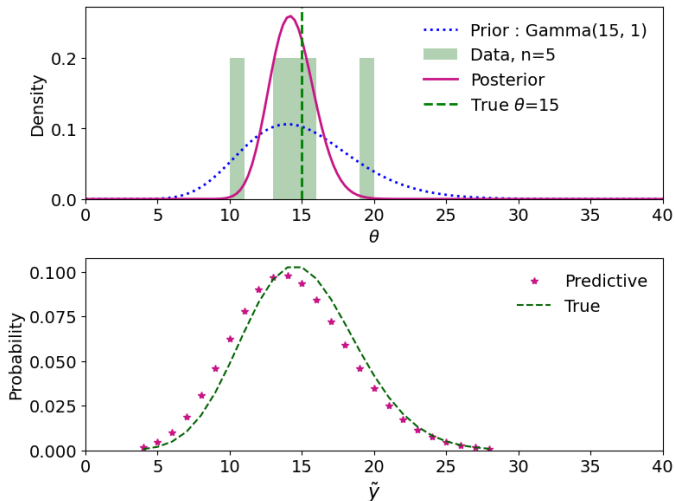
Predictive Posterior



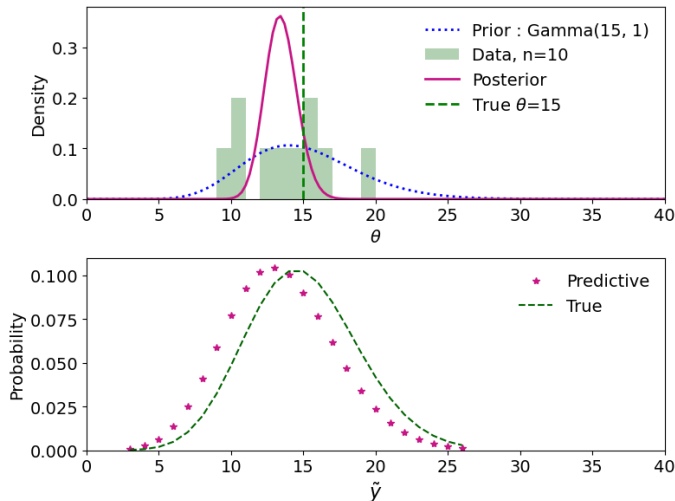
Predictive Posterior



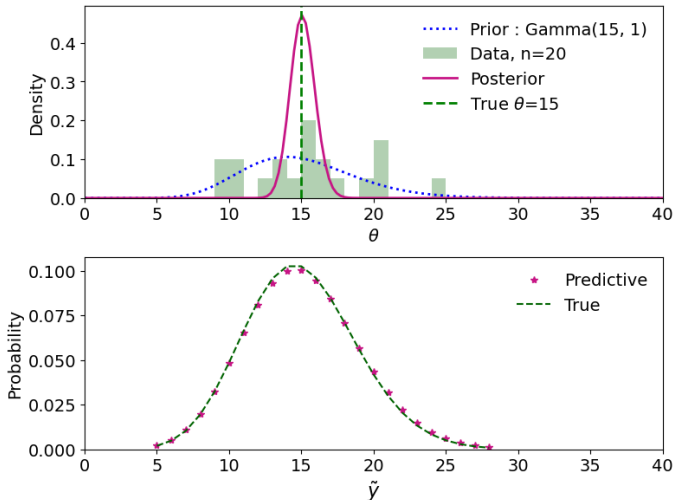
Predictive Posterior



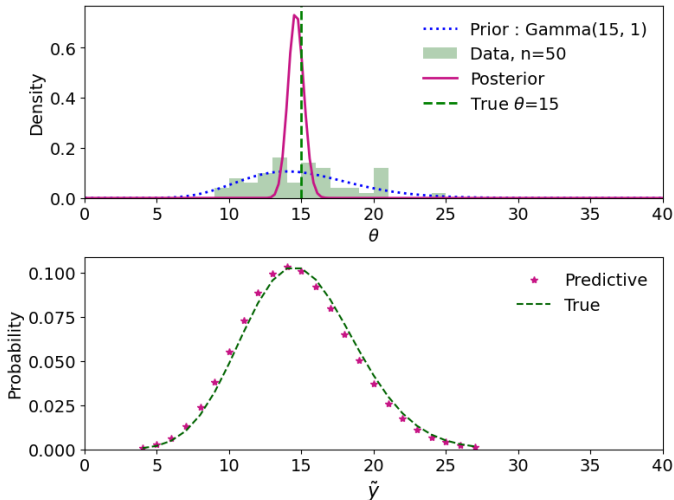
Predictive Posterior



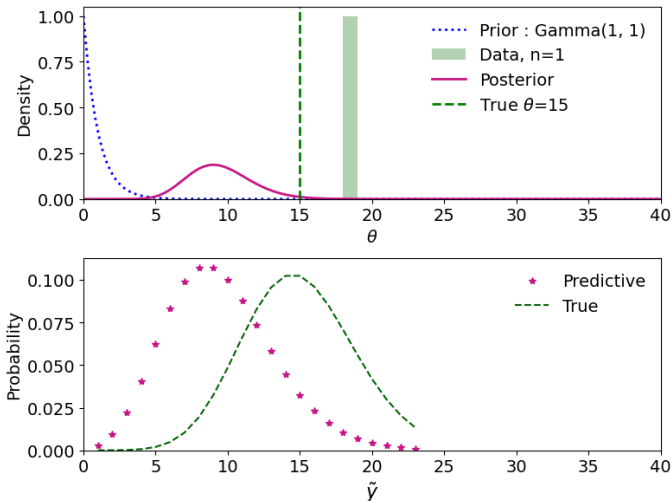
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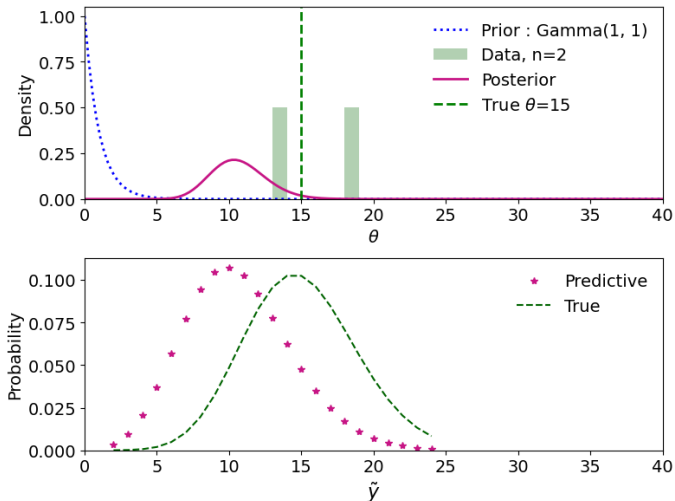
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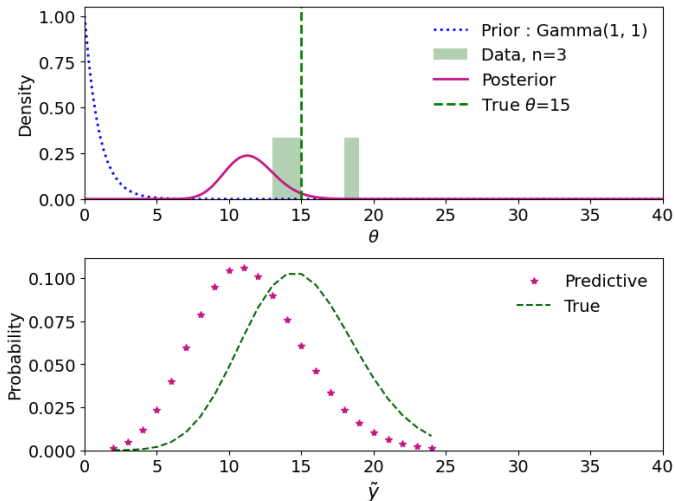
Predictive Posterior with a daft prior



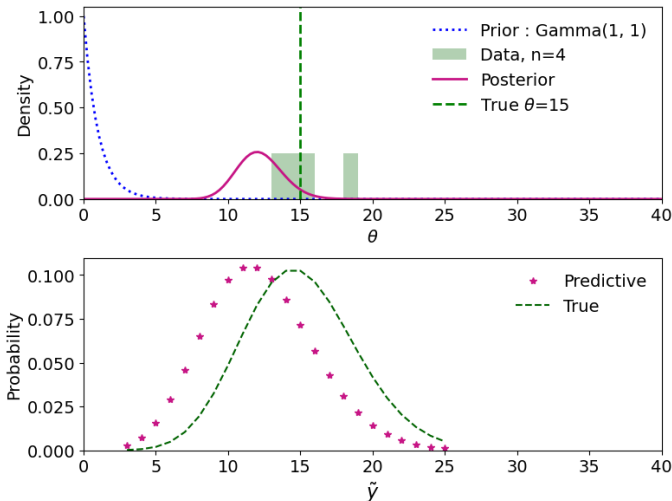
Predictive Posterior with a daft prior



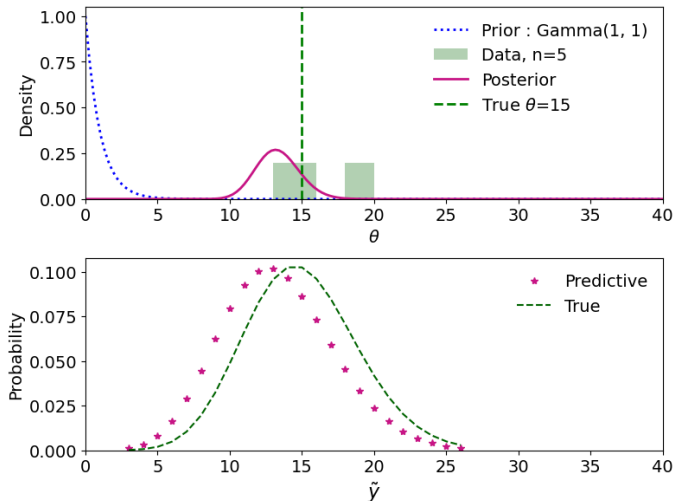
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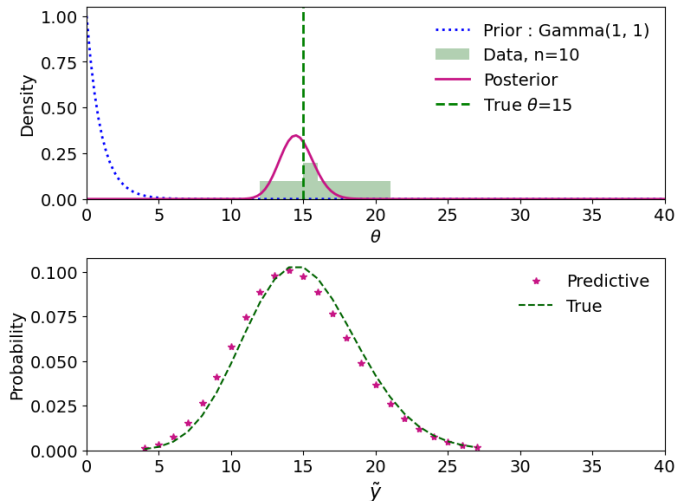
Predictive Posterior with a daft prior



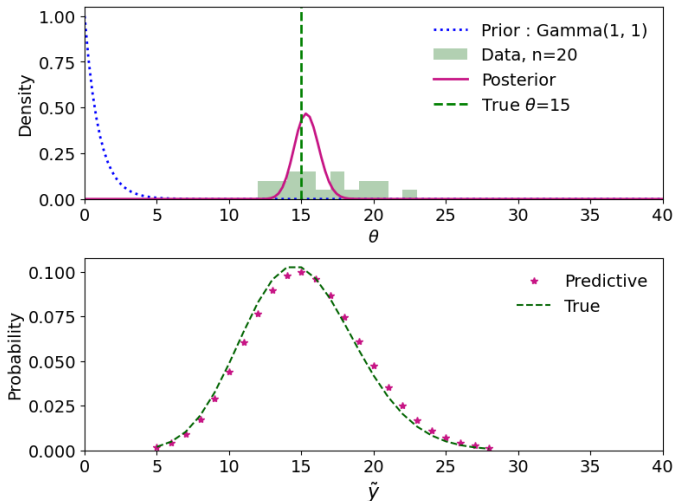
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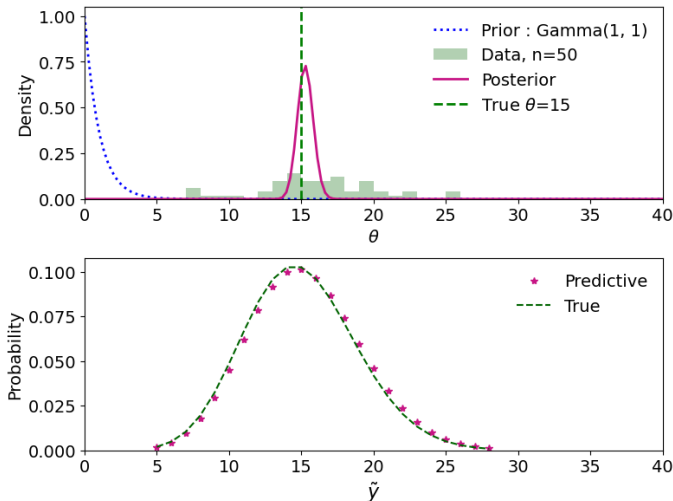
Predictive Posterior with a daft prior



Predictive Posterior with a daft prior



Predictive Posterior with a daft prior



It's not Magic, it's Bayesian Updating.

The use of a Prior allows us to preserve the Uncertainty associated with measurements, particularly when N is small.

The use of a Conjugate Prior makes the inference transparent and the analysis easy.

If the Posterior integral is not a nice simple one, we can take samples from it using eg Markov Chain Monte Carlo, which we will look at next week.

Thanks For Your Time

Fin.